

No. of Functional Features Included in the Solution

Date:	3 October 2026	Team ID:	unm1611MADHANRAJ
Project Name:	AI FitTrack API	Maximum Marks:	5 Marks

Functional Features Overview:

This document inventories all 10 functional features developed and verified in the AI FitTrack API solution, detailing their operational purpose, module location, completion status, and mark contributions.

S.No	Feature Name	Feature Description	Module / File	Status	Marks
1	User Registration	Registers new account with bcrypt password encryption (10 salt rounds) and duplicate email check.	authController.js	Done	0.5
2	User Login & JWT	Authenticates email and password, returning signed HMAC-SHA256 JWT access token.	authController.js	Done	0.5
3	User Profile	Returns authenticated user identity and timestamp information excluding password hash.	authController.js	Done	0.5
4	Add Workout	Logs new workout specifying category, duration in minutes, calories burned, and date.	workoutController.js	Done	0.5
5	View All Workouts	Retrieves all workouts logged by authenticated user sorted chronologically (descending).	workoutController.js	Done	0.5

Subtotal Features (Part 1): 5 Features Completed (2.5 / 2.5 Marks)

No. of Functional Features — Category Breakdown & Endpoints

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S.No	Feature Name	Feature Description	Module / File	Status	Marks
6	View Workout by ID	Fetches specific workout entry by MongoDB ObjectId with strict user-ownership validation.	workoutController.js	Done	0.5
7	Update Workout	Modifies workout duration, category, or calories; enforces user ownership validation.	workoutController.js	Done	0.5
8	Delete Workout	Removes workout document from database; verifies user ownership prior to deletion.	workoutController.js	Done	0.5
9	Multi-Criteria Search	Dynamic regex search across workoutName, category, and date formats (?q=keyword).	workoutController.js	Done	0.5
10	AI Workout Plan	Generates personalized workout routines using Google Gemini 1.5 Flash AI model.	aiController.js & geminiService.js	Done	0.5

Total Functional Features: 10 Features Completed (5.0 / 5.0 Marks — 100% Compliant)

Feature Category Breakdown:

S.No	Category	Features Count	Concrete Implementation in AI FitTrack API
1	Authentication & Identity	3 Features	Registration, Login, and Profile retrieval with bcrypt and JWT.
2	Workout Management CRUD	5 Features	Create, Read All, Read by ID, Update, and Delete workouts with user scoping.
3	Search & Discovery	1 Feature	Multi-criteria regex query matching workoutName, category, and date.
4	Generative AI Intelligence	2 Features	Gemini 1.5 Flash AI workout plan recommendation & fitness progress insights.